



Optical wireless positioning by modulation of the optical source orientation

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Abstract: Positioning for optical wireless communications (OWC) has been the subject of numerous theoretical and experimental works in recent years, proposing various techniques for achieving sub-10 cm positioning errors. However, most of these techniques require multiple optical sources and/or multiple photoreceivers. In this paper, we take advantage of the growing interest in beam-steering techniques for OWC to propose a new positioning technique based on the possibility of changing the orientation of a single optical source to enable the two-dimensional positioning of a receiver equipped with a single photodetector, whatever its orientation and the optical power emitted. We detail the theoretical principles of this new method and then validate its operation, first using simulations to extract the main behavior, then experimentally on one dimension. The results obtained show that, in addition to the aforementioned advantages, the proposed method enables positioning with sub-10 cm errors, making it an interesting positioning technique for OWC with beam steering.

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1. Introduction

In the world of optical wireless communications (OWC), directed beams using directional emission patterns have been used for several years for point-to-point links. The related developments have been pushed by the free space optics (FSO) domain [1], that aims at high data rate connections with lasers in many fields like space applications, inter-buildings communications, and unnamed aerial vehicle transmissions for example [2]. In addition, with recent deployments of visible light communications (VLC) and light fidelity (LiFi), directional beams are widely explored in all kinds of OWC systems [3–5]. Advantages are numerous, like minimizing multipath dispersion, potential extension of coverage area, dynamic network reconfiguration with minimization of handover, and high improvement of signal-to-noise ratio (SNR).

Nevertheless, several challenges remain to be solved to reach such performance. In particular, the emitting antenna has to be able to perform beamsteering, i.e. to steer its optical beam toward the desired user. At the same time, the user equipment (UE) better adapt its field of view (FOV) to capture the incoming light signal of interest while minimizing the surrounding noises and disturbances. These processes need to be as agile as possible to track the target in a mobile configuration. Several beamsteering techniques have therefore been proposed for OWC, including gimbals, micro-mirrors, liquid crystals, and phased arrays [1], depending on the steering range and speed that are necessary.

Although each method offers its own advantages and limitations, one common challenge is that the emitter has to know accurately the position of the targeted user. This position can be estimated independently of the data communication function, but a growing body of research is

now exploring the combination of the two functions in an integrated sensing and communication (ISAC) approach [6–8]. In any case, a lot of optical wireless positioning (OWP) methods have been developed using photodetector or camera as light receivers [9–16], sometimes inspired from the radio frequency (RF) field [17].

A first group of methods, often referred to as proximity, relies on emitters transmitting each a dedicated code or frequency. Receivers are then located by decoding these beacons, but the resolution remains quite poor and can only be enhanced by increasing the density of optical sources. A second group of methods consists in measuring the time of arrival (TOA), time difference of arrival (TDOA) or phase of arrival (POA) [18–20] of one or more signals between the moment they are transmitted and received, in order to determine their propagation distance using the speed of light. However, these techniques require very precise measurements of short times to achieve good positioning performance.

The third group of commonly used OWP methods is based on the angle of arrival (AOA) measurement combined with triangulation algorithms [21–36], whereas the fourth and probably the main group of methods relies on the measurement of the received signal strength (RSS) combined with trilateration [37–44]. The main advantage of the latter lies in its ease of use, as the optical receiver necessarily includes analog-to-digital converters allowing to compute the signal power. The direct measurement of the AOA is more complex, but can be done in practice from the RSS of different signals, which is why hybrid RSS/AOA-based positioning has also been studied over the past years [45,46]. In parallel, machine learning algorithms have been extensively developed to improve the positioning performance [13,14].

Table 1 summarizes the main features of state-of-the-art work on the use of RSS and AOA techniques for OWP. While it can be seen that cm-level accuracy is the norm, it can also be observed that whatever the employed method, it is necessary to use several transmitters [21–24,28,30,40,43,46], several receivers [26,33,34,42,46] and even complementary sensors such as camera [25,29,32,35,36], magnetic sensor [37] or inertial sensors [27,38,39], which may induce space constraints, particularly if several receivers are needed in a UE to be used in embedded IoT systems for example. In addition, these methods usually require knowledge of the transmitted power and are sensitive to the orientation of the receiver, a set of drawbacks that we propose to overcome in this work.

Table 1. Comparison between the existing literature on OWP and our work. ^a

Reference	Year	2D 3D	Positioning technique	Setup	Type	Accuracy (type)	Testbed size
[33]	2020	2D	AOA	1 LED / 4 SC with aperture	Experiment	3.1 cm (mean)	40 × 40 cm ²
[40]	2021	3D	RSS	4 tilted LED / 1 PD	Simulation	1.3 cm (90%)	6 × 6 × 3 m ³
[46]	2022	2D	Hybrid AOA/RSS	4 LED / 4 PD with aperture	Experiment	16.6 cm (95%)	3 × 3 m ²
[35]	2023	3D	AOA with LMD	1 LED / 1 camera	Experiment	22 cm (90%)	1.84 × 1.84 × 1.96 m ³
[44]	2024	2D	AOA	1 LED / 1 rotatable PD	Simulation Experiment	11.3 cm (92%) 0.9 cm (mean)	5 × 5 m ² 50 × 50 cm ²
This work	2024	2D	Hybrid AOA/RSS	1 rotatable LED / 1 PD	Simulation Experiment	5.4 cm (90%) 0.9 cm (mean)	1 × 1 m ² 3.25 m (1D)

^a 1D/3D: One/three-dimensional, SC: Solar cell, LMD: Luminance distribution model.

In a recent study, Wang *et al.* [44] also went in this direction, associating a single optical source with a receiver equipped with a photodetector which orientation can be modulated. By applying four different rotations to this receiver, it is then possible to perform RSS-based two-dimensional (2D) positioning with cm-level accuracy, without the need to know the transmitted optical power. However, the use of such a receiver can be restrictive in practice, particularly in terms of size and complexity of implementation.

With the recent progresses in beamsteering, it is now easily possible to modulate the orientation of the optical source. Next generation of optical antennas will therefore be similar to RF antennas that enable fine phase tuning to build specific patterns of emission [47]. From a positioning perspective, modulating the orientation of the optical antenna can then replace the need for separate sources.

In this paper, we exploit the possibility provided by beamsteering to modulate the orientation of a source to propose a fully new hybrid RSS and AOA positioning method, where only one pair of emitter-receiver is needed for 2D positioning. Note, however, that this system could be extended to three-dimensional positioning of the receiver by adding a second emitter. Furthermore, the proposed method has the advantage of not depending on either the transmitted power or the orientation of the receiver.

After outlining the operating principles of this method in Section 2, we study its functioning, advantages and limitations with the help of simulations in Section 3. We then propose an experimental validation of this method on one dimension in Section 4, before drawing the main conclusions from this work and identifying relevant future works in Section 5.

Notations: In this paper, a variable in normal font (e.g. x) is a scalar whereas a variable in bold font (e.g. $\mathbf{x}$ or $\mathbf{X}$) is a vector or a matrix. By extension, the latter notation $\mathbf{X}$ may also refer to a point in the 3D space, identified by its coordinates x , y and z in a given Cartesian reference frame noted $\mathcal{F}$, under the form $\mathbf{X} = [x, y, z]_{\mathcal{F}}$. Since only a single reference frame is here considered, this notation is simplified as follows $\mathbf{X} = [x, y, z]$. The subscripts t and r refer respectively to variables related to the transmitter and receiver, whereas the indices i, j and k are mute indices referring, unless specified otherwise, to the different orientations of the transmitter.

2. Working principles of the proposed technique

2.1. Assumptions and modeling

2.1.1. Geometrical modeling

The role of the system studied here is to estimate, in a room of dimensions $L \times W \times H$, the position of an UE equipped with at least one photodetector (PD), using an access point (AP) equipped with at least one optical source whose orientation can be modulated. Such a system can therefore be geometrically modelled as in Fig. 1.

The center of the optical source of the AP is placed at point $\mathbf{T}$, defined as the origin of the main frame $\mathcal{F} = (\mathbf{T}, \mathcal{B})$ where $\mathcal{B} = \{\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z\}$ is an orthonormal basis of $\mathbb{R}^3$. On its side, the center of the UE's PD is placed at a point $\mathbf{R}$ with unknown coordinates $[x, y, z]^T$ (with $z < 0$, so that the UE remains inside the room and in a different plane as the AP). When they are in line-of-sight (LOS), the optical source and the photodetector are thus separated by a distance d along the unit vector $\mathbf{v}_{tr}$, both defined as:

$$d = \|\mathbf{R} - \mathbf{T}\| = \sqrt{x^2 + y^2 + z^2}, \quad (1)$$

$$\mathbf{v}_{tr} = \frac{\mathbf{R} - \mathbf{T}}{\|\mathbf{R} - \mathbf{T}\|} = \frac{1}{d} \cdot [x, y, z]^T. \quad (2)$$

At each instant, the optical source and the PD are oriented in independent directions defined respectively by the unit vectors normal to their surfaces $\mathbf{n}_{t,i}$ and $\mathbf{n}_r$. The positioning method proposed here is then based on modulating the orientation of the optical source using any optical

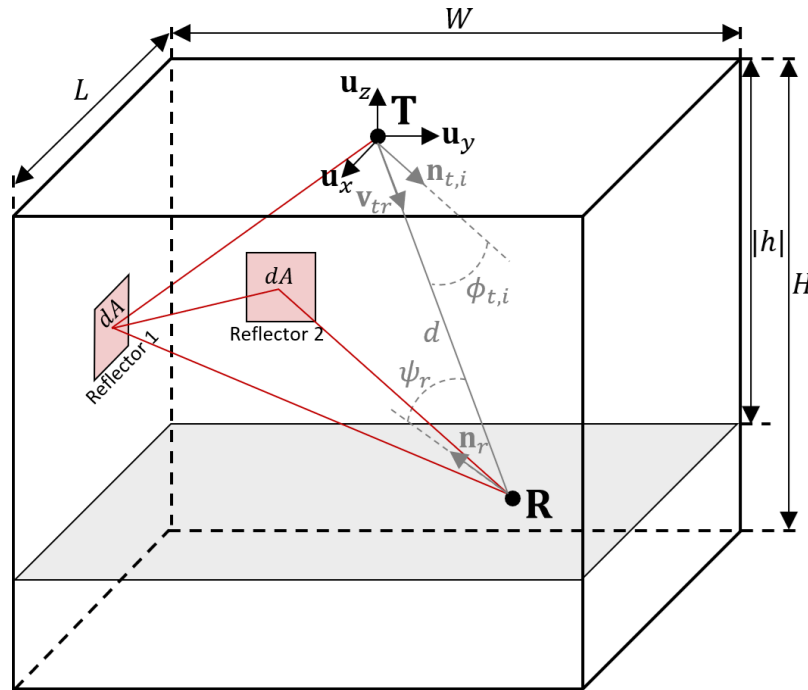


Fig. 1. Geometric configuration of the proposed system and notations (at the exception of the polar angle $\theta_{t,i}$, not represented for the sake of readability).

beamsteering solution that would not change the beam pattern of the source. Detailing these solutions is out of the scope of this article, although an example will be provided in Section 4.1. Here, we will thus model the optical beamsteering system abstractly, as a device able to modify dynamically the source orientation, i.e. the vector $\mathbf{n}_{t,i}$. The mute index i therefore denotes the multiple possible orientations of this transmitter.

In the rest of this paper, we will assume that:

$$\mathbf{n}_{t,i} = [a_i, b_i, c_i]^T, \quad (3)$$

with $\sqrt{a_i^2 + b_i^2 + c_i^2} = 1$ since $\mathbf{n}_{t,i}$ are unit vectors. It is then clear from Fig. 1 that the direction of emission $\mathbf{v}_{tr}$ from the AP towards the UE is not necessarily the same as the direction of orientation of the AP $\mathbf{n}_{t,i}$. Given the definitions of these two vectors, the angle between both directions, called angle of emission (or angle of irradiance), is such that:

$$\cos(\phi_{t,i}) = \mathbf{v}_{tr} \cdot \mathbf{n}_{t,i} = \frac{a_i x + b_i y + c_i z}{d}, \quad (4)$$

where the operator ‘ $\cdot$ ’ is the scalar product between two vectors. Similarly, note that the direction of incidence on the UE, defined by the vector $-\mathbf{v}_{tr}$ and its direction of orientation, defined by the unit vector $\mathbf{n}_r$ normal to its sensitive area, may be different. In such a case, they form an angle noted ψ_r which, however, does not affect our method.

2.1.2. Physical modeling

We now assume that the optical source used on the AP side is oriented in a known direction i (i.e. following the known support vector $\mathbf{n}_{t,i}$, as represented in Fig. 1), and that it outputs a constant average optical power $P_{t,i} = P_t$ with a known but generic beam pattern $R_t(\phi_{t,i}, \theta_{t,i})$,

where $\phi_{t,i}$ and $\theta_{t,i}$ are respectively the polar and azimuth angles with reference to $\mathbf{n}_{t,i}$. In such a case, the optical signal collected by the PD on the UE side is the sum of the signal received after direct LOS propagation with all the non-LOS (NLOS) signals reaching the UE after multiple reflections. Assuming the signal bandwidth is small enough (i.e a few tens of MHz), then the frequency responses H_{LOS} and H_{NLOS} of the LOS and NLOS paths respectively do not vary with the frequency, so the received signal power $P_{r,i}$ is [48]:

$$P_{r,i} = [H_{\text{LOS}} + H_{\text{NLOS}}] P_t. \quad (5)$$

Under this assumption, we can then express H_{LOS} considering that the optical power received from the LOS path will be equal to the optical power emitted P_t , attenuated by: 1) the pattern of emission $R_t(\phi_{t,i}, \theta_{t,i})$ depending on the direction of emission, i.e. the angles $\phi_{t,i}$ and $\theta_{t,i}$, 2) the LOS propagation distance d , and 3) the effective sensitive surface $A_{r,\text{eff}}(\psi_r)$ which varies as a function of the orientation ψ_r of the receiver with respect to the incident LOS ray. In other words, and according to [48], we can say that :

$$H_{\text{LOS}} = \frac{R_t(\phi_{t,i}, \theta_{t,i}) A_{r,\text{eff}}(\psi_r)}{d^2}. \quad (6)$$

At the same time, H_{NLOS} can be expressed as the infinite sum of propagation losses induced by the various possible paths between transmitter and receiver, which can nevertheless be reduced to the finite matrix product proposed in (28) of [49], but not detailed here for the sake of readability.

As we will see in Section 2.2, the proposed positioning method requires knowledge of the optical power received from different directions of emission i . In practice, these values are much easier to measure indirectly from the electrical signals produced by the PD, rather than directly, which would require the use of cumbersome additional equipment. At first order, the instantaneous photocurrent $I_{r,i}(t)$ output by the PD depends indeed on its instantaneous incident optical power $P_{r,i}(t)$ according to the relation:

$$I_{r,i}(t) = \gamma_r P_{r,i}(t) + n(t), \quad (7)$$

with γ_r the PD responsivity, in A/W, and $n(t)$ the noise added by the PD, usually modelled as additive white Gaussian noise (AWGN) [48]. An estimate $\hat{P}_{r,i}(t)$ of $P_{r,i}(t)$ can therefore be obtained from $I_{r,i}(t)$ by simply dividing it with γ_r which, after introduction of (5), gives:

$$\hat{P}_{r,i}(t) = \frac{I_{r,i}(t)}{\gamma_r} = [H_{\text{LOS}} + H_{\text{NLOS}}] P_t(t) + \frac{n(t)}{\gamma_r}. \quad (8)$$

This equation clearly highlights two fundamental elements for our positioning technique. On the one hand, it shows that the received optical power that can be estimated depends directly on H_{LOS} and therefore on the distance d between transmitter and receiver, which ultimately provides crucial information for determining the position of the receiver. On the other hand, it shows that this estimate is biased by at least two sources of error: firstly, AWGN noise, and secondly, its dependence via H_{NLOS} on multipath and therefore on distances other than d . With these elements in mind, we can now enter the details on the proposed positioning method.

2.2. Positioning by modulation of the source orientation

2.2.1. General case

Let an UE of unknown coordinates $\mathbf{R} = [x, y, z]^T$ and an AP equipped with an optical source $\mathbf{T}$ which orientation can be successively changed from direction i to j . First, we assume that the NLOS components of the optical signal received by the UE are negligible, and that the UE's PD

does not introduce any AWGN. In such a case, the estimated received optical power $\hat{P}_{r,i}$ is thus equal to the optical power actually received $P_{r,i}$, so we can deduce from (5) and (6) that:

$$\frac{P_{r,i}}{P_{r,j}} = \frac{R_t(\phi_{t,i}, \theta_{t,i})}{R_t(\phi_{t,j}, \theta_{t,j})}, \quad (9)$$

which is remarkable in that it only depends on the angles of emission $\phi_{t,i}$ and $\theta_{t,i}$, defined themselves by the relative position between the transmitter and the receiver, i.e. by the coordinates x , y and z of the receiver. At the same time, this ratio depends neither on the optical power emitted nor on the orientation of the receiver.

We can therefore conclude that it should be possible to find the receiver's coordinates from one or more ratios of this type. Note that this principle is valid not only for optical sources, which is the case of interest for us here, but also for other types of sources such as RF antennas. Note, in addition, that if we want to go further in our development, a more explicit form of beam pattern is needed.

2.2.2. Case of a Lambertian source (x , y and z unknown)

Let us now assume that the optical source is following a generalized Lambertian pattern of order m_t , which is a very common case in OWC [48]. This means that the pattern of emission is now symmetrical around the optical axis of the source, i.e. only depends on $\phi_{t,i}$ according to a cosine law of order m_t . In other words, the LOS channel gain H_{LOS} can be expressed as [48]:

$$H_{\text{LOS}} = \begin{cases} \frac{A_r(m_t+1)}{2\pi d^2} \cos^{m_t}(\phi_{t,i}) \cos(\psi_r) \\ \text{if } 0 \leq \phi_{t,i} \leq \frac{\pi}{2} \text{ and } 0 \leq \psi_r \leq \Psi_c, \\ 0 \text{ otherwise,} \end{cases} \quad (10)$$

where the Lambertian order of emission, m_t , is defined as $m_t = -\log 2 / \log (\cos \Phi_{1/2})$ with $\Phi_{1/2}$ the semi-angle at half-power of the source, where A_r is the sensitive area of the reception PD and where Ψ_c is its field-of-view (FOV).

By combining (5) with (10) while deliberately neglecting the NLOS channel gain, (9) can then be re-expressed as follows:

$$\frac{P_{r,i}}{P_{r,j}} = \left(\frac{\cos \phi_{t,i}}{\cos \phi_{t,j}} \right)^{m_t}. \quad (11)$$

Using in addition (4), this equation can be rewritten as:

$$\frac{P_{r,i}}{P_{r,j}} = \left(\frac{a_i x + b_i y + c_i z}{a_j x + b_j y + c_j z} \right)^{m_t}, \quad (12)$$

which clearly shows the dependence of the ratio of received powers with the receiver's coordinates, and not with the transmitted optical power or the orientation of the receiver. Note that similar equations would be obtained with any optical source having a symmetrical pattern around its optical axis (i.e. independent of $\theta_{t,i}$), so that the proposed method could also be used with Gaussian laser sources or even imperfect Lambertian sources, as the one later used in Section 4.

Let us now introduce the factor K_{ij} , defined from (12) as:

$$K_{ij} = \left(\frac{P_{r,i}}{P_{r,j}} \right)^{\frac{1}{m_t}} = \frac{a_i x + b_i y + c_i z}{a_j x + b_j y + c_j z}. \quad (13)$$

These factors are in practice known as they can be calculated from the estimates of the received optical powers in the different directions of emission. By reformulating (13), we can then obtain

the following simple linear equation with unknowns x , y and z :

$$(a_i - K_{ij}a_j)x + (b_i - K_{ij}b_j)y + (c_i - K_{ij}c_j)z = 0. \quad (14)$$

In order to find the values of these unknowns, it is therefore necessary to get other linear equations of similar form, which can be achieved by orienting the transmitter in other directions. If we now consider one additional random orientation k , and the resulting ratios K_{ij} , K_{jk} and K_{ik} , we can express the corresponding Eqs. (14) in the following matrix form:

$$\mathbf{A}\mathbf{R} = \mathbf{0}_{3,1}, \quad (15)$$

with $\mathbf{R} = [x, y, z]^T$, $\mathbf{0}_{3,1} = [0, 0, 0]^T$ and:

$$\mathbf{A} = \begin{bmatrix} a_i - K_{ij}a_j & b_i - K_{ij}b_j & c_i - K_{ij}c_j \\ a_j - K_{jk}a_k & b_j - K_{jk}b_k & c_j - K_{jk}c_k \\ a_i - K_{ik}a_k & b_i - K_{ik}b_k & c_i - K_{ik}c_k \end{bmatrix}. \quad (16)$$

This system has a unique solution if and only if the determinant of $\mathbf{A}$ is non-zero. However, as shown in the [Appendix](#) and under the current assumptions, this determinant is zero whatever the orientations of the source. It is therefore necessary to formulate additional hypotheses in order to make this system solvable.

2.2.3. Case of a Lambertian source (z known)

Several different assumptions could be used to make the system formed by (15) and (16) solvable, but we focus here on the case where the vertical distance z between the AP and the UE is known, e.g. $z = h$, with $h < 0$. Such an assumption may seem restrictive but is in practice verified in many cases (e.g. mobile robot in a plane) or can at least be validated with additional sensors.

In this case, Eq. (15) has only two unknown variables left (x and y) and therefore only needs two equations to be solved, which can for example be obtained with the ratios K_{ij} and K_{ik} . Solving the resulting system by simple substitution gives eventually:

$$x = \frac{(K_{ik}b_k - b_i)(c_i - K_{ij}c_j) - (K_{ij}b_j - b_i)(c_i - K_{ik}c_k)}{(K_{ik}b_k - b_i)(K_{ij}a_j - a_i) - (K_{ij}b_j - b_i)(K_{ik}a_k - a_i)}h, \quad (17)$$

$$y = \frac{(K_{ik}a_k - a_i)(c_i - K_{ij}c_j) - (K_{ij}a_j - a_i)(c_i - K_{ik}c_k)}{(K_{ik}a_k - a_i)(K_{ij}b_j - b_i) - (K_{ij}a_j - a_i)(K_{ik}b_k - b_i)}h. \quad (18)$$

In other words, by orienting the optical source of the AP in three different directions, and assuming that the z coordinate of the UE is known, it is then possible to estimate its remaining coordinates x and y , regardless the transmitted optical power and the orientation of the UE.

It may be noted, however, that Eqs. (17) and (18) are only valid as long as the optical power received from the Lambertian source is non-null, i.e. as long as the angle of emission $\phi_{t,i}$ of the source is lower than 90° and/or the angle of incidence is lower than the FOV of the PD. It is also important to note that this position estimate actually neglects the NLOS components of the received optical signals, as well as the AWGN added by the UE's PD, which can therefore have an impact on the positioning error, as we will detail in Section 2.3.

2.2.4. Comparison with RSS and AOA principles

Before examining the sensitivity of our positioning method to various physical parameters, it is interesting to highlight the differences between this method and conventional OWP techniques such as RSS and AOA, by expressing their operation in the same terms as those previously

used. Consequently, let us consider here two different optical sources, indexed i and j , of known coordinates $\mathbf{T}_i = [x_i, y_i, 0]$ and $\mathbf{T}_j = [x_j, y_j, 0]$. We assume they both point to the ground, i.e. $\mathbf{n}_{t,i} = \mathbf{n}_{t,j} = [0, 0, -1]$, while the UE, of position $\mathbf{R} = [x, y, h]$, points to the ceiling, i.e. $\mathbf{n}_r = [0, 0, 1]$.

Since $\mathbf{n}_{t,i}$, $\mathbf{n}_{t,j}$ and $\mathbf{n}_r$ are all collinear, each LOS link between a source and the receiver has equal angles of emission and reception, i.e. $\phi_{t,i} = \psi_{r,i}$ and $\phi_{t,j} = \psi_{r,j}$. This means they also have equal cosines, given in the case of the i -th source by:

$$\cos \phi_{t,i} = \cos \psi_{r,i} = \frac{h}{d_i}, \quad (19)$$

with d_i the LOS distance between this source and the receiver. Using (1), we can express d_i as a function of the source and receiver coordinates, which gives eventually:

$$\phi_{t,i} = \psi_{r,i} = \arccos \left[\frac{h}{\sqrt{(x - x_i)^2 + (y - y_i)^2 + h^2}} \right]. \quad (20)$$

At the same time, the optical powers $P_{r,i}$ and $P_{r,j}$ received respectively from sources i and j while neglecting the NLOS paths can be expressed from (5) and (10), so that the power ratio (11) eventually becomes:

$$\frac{P_{r,i}}{P_{r,j}} = \left(\frac{d_j}{d_i} \right)^{m_t+3}, \quad (21)$$

which, combined with (1), gives:

$$\frac{P_{r,i}}{P_{r,j}} = \left[\frac{(x - x_j)^2 + (y - y_j)^2 + h^2}{(x - x_i)^2 + (y - y_i)^2 + h^2} \right]^{\frac{m_t+3}{2}}. \quad (22)$$

As pointed out in [16], AOA-based positioning relies on the measurement, by the UE, of the angles with which signals coming from one or several sources are received by one or several sensors. In other words, AOA positioning consists in estimating the angles $\psi_{r,i}$ and then deducing with (20) the x and y coordinates of the UE. In practice, this estimation is done using several PDs with specific apertures on top of them [45,46], and/or organized in pair or quadrant [32], or even using cameras [31,35]. In any case, it appears clearly from (20) that at least two angles, e.g. $\psi_{r,i}$ and $\psi_{r,j}$, need to be estimated to retrieve x and y , which means at least two distinct optical sources are necessary for the AOA method to operate properly in 2D [16].

On its side, RSS-based positioning relies on the measurement, by the UE, of the optical power it receives from different and spatially distinct sources. In other words, RSS positioning consists in estimating the powers $P_{r,i}$ and $P_{r,j}$ and then deducing with (22) the x and y coordinates of the UE. Here again, however, these unknowns can only be determined if a second and independent equation is formed, which is not possible with the sources i and j alone. This means a third source k has to be taken into account, so that 2D RSS positioning necessarily requires at least three separate optical sources to operate properly [16].

As a comparison, and conclusion, the method we propose requires just one optical source on the AP side, and one simple PD on the UE side, which is its main advantage compared to the state-of-the-art. This comes eventually from the fact that changing the orientation of a single source is theoretically equivalent to using virtual but distinct optical sources and then applying conventional RSS equations to recover the UE coordinates. Therefore, the strength of our method also comes with an extra cost, which is the need for an optical beamsteering system. However, the growing interest in such technologies to optimize communication performance lets us think that beamsteering will be a major component of future OWC systems, so the positioning method proposed here could well be used to implement OWC-based ISAC systems.

2.3. Sensitivity and limits of the proposed technique

The various equations derived so far show that the accuracy of our positioning method can be influenced by a number of parameters. We have identified at least 11 of these parameters, so that we propose in this section to list them (with numbers '1', '2', '3' etc.) and highlight how they can degrade positioning accuracy.

2.3.1. Incidence of an inaccurate knowledge of h , $P_{r,i}$ and m_t

It is clear from (17) and (18) that the estimation accuracy of the UE's coordinates strongly depends on the accurate knowledge of: 1) the z -coordinate h of the UE, and of the parameters K_{ij} and K_{ik} , which depend themselves on 2) the Lambertian order of the source m_t , and on the received optical powers $P_{r,i}$, $P_{r,j}$ and $P_{r,k}$. First, let us suppose that the z -coordinate of the UE, h , is known with an error ∂h . In such a case, the resulting errors ∂x in x and ∂y in y can be obtained from the partial derivatives of, respectively, (17) and (18) according to h , which, all the other parameters remaining constant, gives:

$$\frac{\partial x}{\partial h} = \frac{(K_{ik}b_k - b_i)(c_i - K_{ij}c_j) - (K_{ij}b_j - b_i)(c_i - K_{ik}c_k)}{(K_{ik}b_k - b_i)(K_{ij}a_j - a_i) - (K_{ij}b_j - b_i)(K_{ik}a_k - a_i)}, \quad (23)$$

$$\frac{\partial y}{\partial h} = \frac{(K_{ik}a_k - a_i)(c_i - K_{ij}c_j) - (K_{ij}a_j - a_i)(c_i - K_{ik}c_k)}{(K_{ik}a_k - a_i)(K_{ij}b_j - b_i) - (K_{ij}a_j - a_i)(K_{ik}b_k - b_i)}. \quad (24)$$

In a second step, let us suppose that the parameters of type K_{ij} are known with an error ∂K_{ij} . From a purely mathematical point of view, the incidence of this error on the estimated position of the UE can be analytically derived from partial derivatives of (17) and (18), which gives in the case of $\partial x / \partial K_{ij}$ the Eq. (25) below. Note that partial derivatives of x and y as a function of any other parameter of type K_{ij} (e.g. K_{ik}) will have a similar form, with a simple change of indices, which is why we only give here:

$$\begin{aligned} \frac{\partial x}{\partial K_{ij}} = & \frac{hc_j(b_i - K_{ik}b_k)}{(K_{ij}a_j - a_i)(K_{ik}b_k - b_i) - (K_{ik}a_k - a_i)(K_{ij}b_j - b_i)} \\ & - \frac{ha_j(K_{ik}b_k - b_i) [(c_i - K_{ij}c_j)(K_{ik}b_k - b_i) - (c_i - K_{ik}c_k)(K_{ij}b_j - b_i)]}{[(K_{ij}a_j - a_i)(K_{ik}b_k - b_i) - (K_{ik}a_k - a_i)(K_{ij}b_j - b_i)]^2}. \end{aligned} \quad (25)$$

From a more practical point of view, we can see from the definition of K_{ij} in (13) that errors in this parameter may result from errors in the Lambertian order of emission m_t and/or errors in the received optical powers $P_{r,i}$ and $P_{r,j}$. The potential incidence of an error in m_t is, on the one hand, not critical. As pointed out in the Section 2.2, the positioning method studied here is operational with any optical source, especially if its emission pattern is symmetrical around its optical axis. Consequently, if the source used is not perfectly Lambertian – and this is often the case with real LED, as we will see in the Section 4.1 – we simply have to replace the Lambertian pattern with a fit of the real pattern in (11) to eventually obtain accurate estimates of x and y .

On the other hand, we understand from (8) that the optical powers estimated are necessarily subject to errors, since they actually include two more sources of degradation of the positioning accuracy: 3) the NLOS optical components of the received signals and 4) AWGN. Both phenomena are neglected to calculate the x and y coordinates in (17) and (18), which is why they may induce positioning errors. The terms $P_{r,i}$, $P_{r,j}$ and $P_{r,k}$ eventually used to perform positioning are, in other words, obtained from signals which SNR is in practice degraded by NLOS propagation paths and noise.

Note that the incidence of these parameters on the estimated x and y could also be studied theoretically through partial derivatives. However, the resulting equations, while analytically interesting, would not be practical to study outside an application context. This is why we have

eventually preferred to explore the incidence of NLOS propagation paths and AWGN through the simulations detailed in Section 3.2, rather than in a purely analytical way.

2.3.2. Incidence of the orientations of the source

It is also clear that (17) and (18) depend on the coefficients $\{a_l, b_l, c_l\}_{l=i,j,k}$ of the three successive orientations $\mathbf{n}_{t,i}$, $\mathbf{n}_{t,j}$ and $\mathbf{n}_{t,k}$ necessary to perform positioning. Therefore, two additional factors may influence the positioning accuracy: 5) the inaccurate knowledge of the source orientations, and 6) the very choice of these orientations.

The influence on x and y of errors in the knowledge of the successive orientations of the source can be studied by the partial derivatives of (17) and (18) as a function of $\{a_l, b_l, c_l\}_{l=i,j,k}$. However, given the form of (17) and (18), these partial derivatives have a form similar to (25) and are therefore not transcribed here. In addition, as this source of error is, like NLOS paths and AWGN, dependent on the application, it will be studied through simulations in Section 3.3.

The incidence of the choice of orientations will also be one of the focus of Section 3.3, but we can already mention that this parameter may have a major incidence on the SNR of the received signal and therefore on the positioning accuracy. For example, if the source is overall pointing in the opposite direction of the UE, we understand that the signal collected from the LOS will be rather low compared to AWGN or NLOS paths, which will translate into a strongly biased estimate of the received optical power, and then of the UE's coordinates.

2.3.3. Incidence of other parameters through the SNR

The choice of the optical source orientations is not the only parameter with a direct incidence on the SNR of the received signals, and therefore potentially on the positioning error. Other parameters that can have a similar impact include 7) the transmitted optical power, 8) the receiver orientation and 9) the sensitive area of the PD used by the UE.

As mentioned in Section 2.2, our positioning method is mathematically independent from these parameters, as they cancel each other out when calculating the ratios (11). In practice, however, these parameters will have an impact on the power of the collected signals. For example, larger transmitted optical power will necessarily result in stronger received signals and thus higher SNR, although the NLOS path might also be stronger. Similarly, if the UE is largely misaligned with the AP, the received signal will get weaker, hence degrading the SNR and thus the positioning accuracy.

Following the same logic, we can identify other parameters that may have an incidence in the 2D localization error of our method, namely: 10) the sensitivity of the PD used by the UE and 11) the position of the UE. Given that most of these parameters are application-dependent, they will be fixed at realistic values for simulations in Section 3 and then will not change, so that their incidence on the positioning accuracy will not be studied, at the exception of the position of the UE which is a critical to obtain a relevant positioning performance indicator.

3. Simulation validation of the proposed method

3.1. Simulation model and parameters

In order to validate the operation of the proposed positioning method, we used MATLAB to model an environment similar to that shown in Fig. 1, which various parameters are listed with their values in Table 2. This environment consists of a room of dimensions $L \times W \times H$, with a ceiling-mounted optical source which orientation $\mathbf{n}_{t,i}$ can be changed. As stated in Section 2.2, three different source orientations are needed for our positioning method to operate. Here, they are set as follows: 1) the source points directly towards the ground ($\mathbf{n}_{t,i} = [0, 0, -1]^T$), 2) it points towards the ground with an angle ω around the axis $\mathbf{u}_x$ ($\mathbf{n}_{t,j} = [0, \sin \omega, -\cos \omega]^T$), and 3) with an angle ω around the axis $\mathbf{u}_y$ ($\mathbf{n}_{t,k} = [\sin \omega, 0, -\cos \omega]^T$). On the UE side, the PD points

towards the ceiling ($\mathbf{n}_r = [0, 0, 1]^T$) while moving in the $\mathbf{u}_x$ and $\mathbf{u}_y$ directions by steps of 10 cm along a reception plane located at a known distance $|h|$ from the source.

Table 2. Simulation parameters and values.

Room parameters	
Parameter	Value
Dimensions ($L \times W \times H$)	4 m $\times$ 4 m $\times$ 2.5 m
Area of the wall reflectors (dA)	100 cm ²
Walls reflectivity (ρ)	0 or 0.6
AP parameters	
Parameter	Value
Coordinates	$\mathbf{T} = [0, 0, 0]^T$
Semi-angle at half-power ($\Phi_{1/2}$)	45°
Transmitted optical power (P_t)	405 mW
Source orientations ($\mathbf{n}_{i,l}$)	$\mathbf{n}_{i,l} = [0, 0, -1]^T$ $\mathbf{n}_{i,j} = [0, \sin \omega, -\cos \omega]^T$ $\mathbf{n}_{i,k} = [\sin \omega, 0, -\cos \omega]^T$
Angle of orientation (θ)	Variable
UE parameters	
Parameter	Value
Coordinates	$\mathbf{R} = [x, y, h]^T$
Height of the UE ($H - h $)	0.85m (i.e. $ h = 1.65$ m)
FOV of the PD (Ψ_c)	85°
Responsivity of the PD (R_{PD})	0.63 A/W
Effective area of the PD (A_{PD})	26.4mm ²
Orientation of the PD ($\mathbf{n}_r$)	$\mathbf{n}_r = [0, 0, 1]^T$

In order to estimate the coordinates of this UE, it is necessary to simulate, for each source orientation, the optical power received after emission at point $\mathbf{T}$ and propagation through the LOS and NLOS paths. For this purpose, we have adopted the frequency-domain OWC channel model proposed by Schulze [49], which was simplified by not taking propagation delays into account. According to this model, the walls, ceiling and floor of the environment are each divided into elementary surface reflectors of area dA , like ‘Reflector 1’ and ‘Reflector 2’ shown in Fig. 1. These reflectors behave like passive Lambertian sources of semi-angle at half power 60° reflecting light rays coming directly from the optical source or from other reflectors with a reflection coefficient ρ either equal to 0 (LOS-only case) or 0.6 (LOS and NLOS case, with a common room reflectivity [50]). Here, we have limited the size of these reflectors to 10×10 cm² in order to model this reflection phenomenon with sufficient finesse, while limiting simulation time, which increases very rapidly as dA decreases.

Finally, in order to evaluate the performance of the UE coordinate estimation, we adopted as our main metric the root mean square (RMS) error threshold at 90%, noted $\varepsilon_{90\%}$, which defines the value below which 90% of the RMS errors on a given reception plane or sub-plane lie. We then studied the evolution of this metric as a function of various parameters, including the reflectivity ρ of the environment, the angle of orientation ω of the source, as well as the AWGN noise level. In the latter case, we varied the SNR threshold, named $\text{SNR}_{90\%}$, above which 90% of the SNRs in the plane or sub-plane under study lie.

3.2. Simulation study of the proposed method

First, we studied the positioning performance of our method according to different sets of parameters, in order to highlight its main characteristics. Figure 2 shows a top view of the reception plane, where the simulated UE positions are marked by black circles and where the positions estimated with our method while neglecting the NLOS paths and considering a very low noise level ($\text{SNR}_{90\%} = 65 \text{ dB}$) are marked by blue dots ('Configuration 1'). In such a case, we can clearly observe a very good agreement between the estimated and actual positions, as also confirmed by the $\varepsilon_{90\%}$ threshold, which is only 0.8 mm. These first results therefore confirm that, in an ideal case, the method proposed here is functional. Note that if we consider, in this configuration, that h is known with an error of 1 cm, then the excess error in $\varepsilon_{90\%}$ is around 1.7 cm which, although not negligible, remains limited.

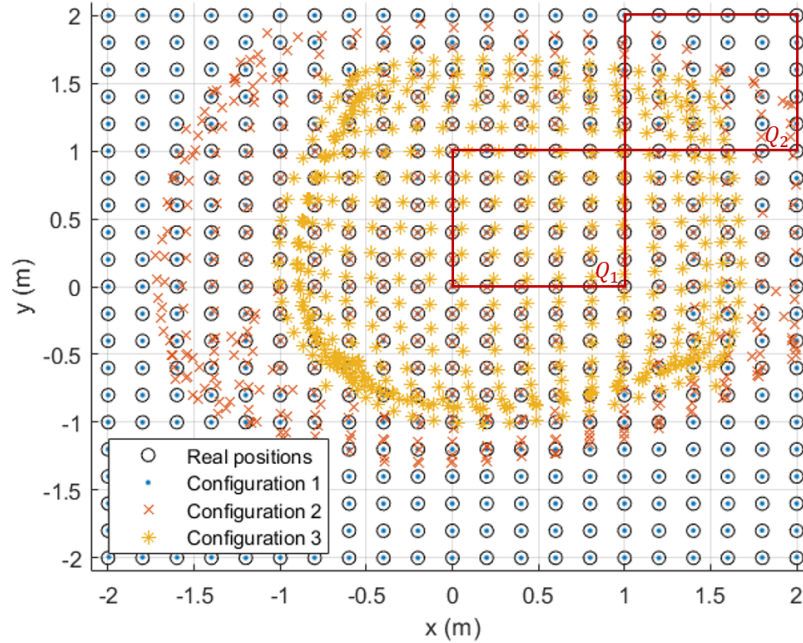


Fig. 2. Top view of the reception plane with the real positions of the UE and its estimated positions under different assumptions (only one position over two is shown to make the figure easier to read).

In practice, however, reflections and noise cannot be underestimated. In Fig. 2, the orange cross-shaped markers therefore correspond to the estimated positions in the case where NLOS paths are still neglected but where the $\text{SNR}_{90\%}$ is now down to 25 dB ('Configuration 2'), whereas the yellow cross-shaped markers correspond to the case where the walls all have a reflection coefficient $\rho = 0.6$, whereas $\text{SNR}_{90\%} = 65 \text{ dB}$ ('Configuration 3'). Note that we have deliberately simulated these two cases separately in order to show that the two corresponding phenomena – reflections and noise – have roughly the same effect on the spatial distribution of the positioning error.

On the one hand, we can now clearly observe positioning errors, which the $\varepsilon_{90\%}$ thresholds allow us to quantify at 102.7 cm over the whole reception plane in 'Configuration 2', and at 113.4 cm in 'Configuration 3'. However, these global values mask the large variation in the RMS error depending on the position of the UE, which we can clearly see when comparing, for example, the estimated positions on the sub-planes $Q_1 = \{(x, y) | x \in [0, 1], y \in [0, 1]\}$ and $Q_2 = \{(x, y) | x \in [1, 2], y \in [1, 2]\}$, highlighted with the red squares in Fig. 2. In such sub-planes,

we find that $\varepsilon_{90\%}$ goes respectively down to 1.0 cm and 42.2 cm in ‘Configuration 2’ whereas it reaches 7.2 cm and 58.7 cm in ‘Configuration 3’.

These large spatial differences first stem from the fact that, whatever the orientation of the optical source, both AWGN and reflections contribute to increasing the value of the estimated received optical power $\hat{P}_{r,i}$ compared to the noiseless and LOS-only received optical power, as suggested by (8). Consequently, these two phenomena contribute to a variation in the coefficients of type K_{ij} , which eventually leads to errors in the estimated coordinates, as highlighted by (25).

In the case of ‘Configuration 2’, this error then gets larger as the the optical power received decreases, whatever the source orientation, since AWGN is considered constant. As a consequence, in the Q_2 sub-plane, the $\text{SNR}_{90\%}$ drops to 16.8 dB, whereas it reaches 28.0 dB in the Q_1 sub-plane. In the case of ‘Configuration 3’, it is the share of NLOS paths in the received optical power that ultimately explains the spatial variation in RMS error. In the Q_1 sub-plane, this share reaches a maximum of 26.5% of the total received optical power, whereas in the Q_2 plane, it reaches 76.2%, thus introducing a much greater error. In other words, the positioning method we propose is all the more reliable when the receiver is located in an area well illuminated by the LOS path, whatever the three successive source orientations required for position estimation. The choice of these orientations is therefore critical to improving positioning performance.

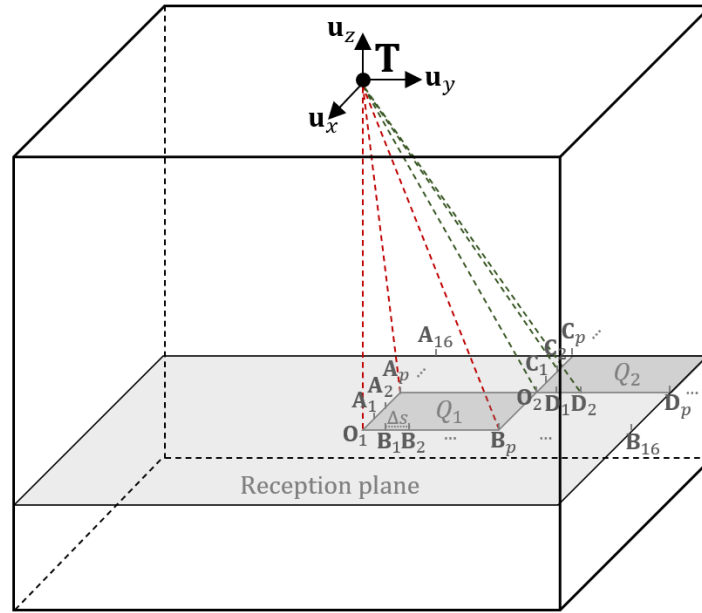
3.3. How to choose the different orientations

To illustrate the incidence of the source orientations $\mathbf{n}_{t,i}$, $\mathbf{n}_{t,j}$ and $\mathbf{n}_{t,k}$ on the positioning performance, we calculated again the RMS error $\varepsilon_{90\%}$ over the sub-planes Q_1 and Q_2 , but this time varying the angle ω defining the directions of $\mathbf{n}_{t,j}$ and $\mathbf{n}_{t,k}$ while keeping $\mathbf{n}_{t,i} = [0, 0, -1]$ (i.e. pointing toward $\mathbf{O}_1$). To do this, we defined two sets of target points on the receiver plane, distributed at regular intervals Δs , respectively along the axes $\mathbf{u}_x$ and $\mathbf{u}_y$. In other words, as represented in Fig. 3(a), we defined the points $\mathbf{A}_p = [-s_p, 0, h]$ and $\mathbf{B}_p = [0, s_p, h]$, with $s_p = p\Delta s$, $\Delta s = 12.5$ cm and $m \in \{1, \dots, 16\}$, i.e. $s_p \in [12.5 \text{ cm}, 200 \text{ cm}]$. Then, for a given p , we set the second and third orientations of the source such that they were pointing towards the points $\mathbf{A}_p$ and $\mathbf{B}_p$ respectively, i.e. $\mathbf{n}_{t,j} = \frac{\mathbf{A}_p - \mathbf{T}}{\|\mathbf{A}_p - \mathbf{T}\|}$ and $\mathbf{n}_{t,k} = \frac{\mathbf{B}_p - \mathbf{T}}{\|\mathbf{B}_p - \mathbf{T}\|}$ (see for example the p -th set of orientations in red dashed lines in Fig. 3(a)). In parallel, we took into account NLOS paths, with $\rho = 0.6$, but neglected the AWGN (i.e. $\text{SNR}_{90\%} = +\infty$) to ease performance comparison between both sub-planes.

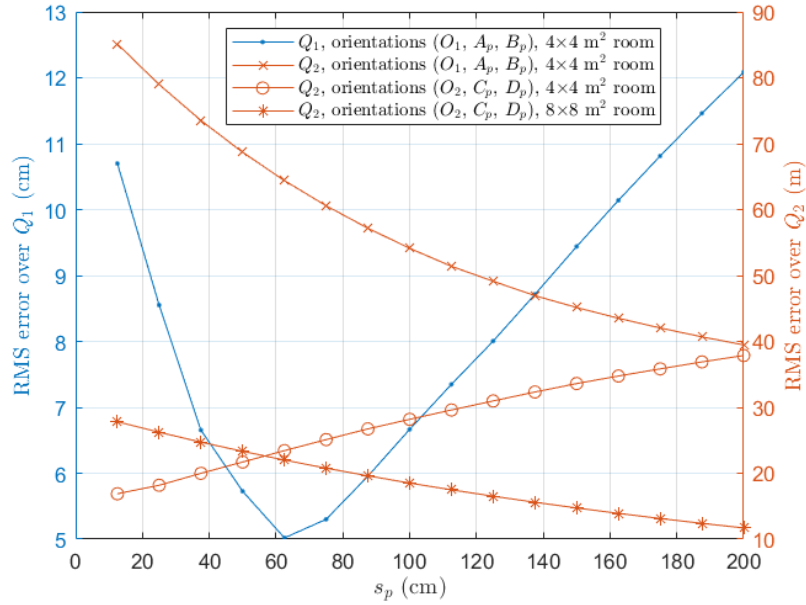
Figure 3(b) shows the evolution, under these conditions and as a function of the source orientation through the variable s_p , of the RMS positioning error $\varepsilon_{90\%}$ on sub-planes Q_1 (plain blue curve, left y-axis) and Q_2 (orange curve with cross markers, right y-axis). In the case of Q_1 , we observe that this error initially decreases with s_p , before reaching a minimum value of 5.0 cm when $s_p = 62.5$ cm (i.e. $\omega = 20.7^\circ$), and then increases. Conversely, the error on Q_2 continuously decreases as s_p increases, but in the best case only reaches 39.5 cm.

Two main conclusions can therefore be drawn from these results. Firstly, we can see that despite reflections, and without any specific technique to mitigate their effects, the proposed technique may still enable, in certain conditions, positioning with a good sub-10 cm resolution. Secondly, it appears clearly that the set of orientations minimizing the error on a given sub-plane depends on the position of this sub-plane relative to the source. Note here that we have in all cases set the first orientation such that the source points directly towards the ground, a choice that may not be optimal. A much more exhaustive study, outside the scope of this article, would therefore be needed to define these optimal orientations.

To illustrate this second conclusion, however, we again studied the case of the Q_2 sub-plane, this time with orientations which, although probably non-optimal, theoretically allow for an increase in received LOS optical power. We have thus defined the first orientation $\mathbf{n}_{t,i}$ so that the source points towards the lower left corner of the Q_2 sub-plane (i.e. towards the point $\mathbf{O}_2 = [1, 1, h]$) and the other two orientations $\mathbf{n}_{t,j}$ and $\mathbf{n}_{t,k}$ such that the source points respectively



(a) Illustration of the pointing strategies used to improve positioning performance.



(b) Evolution of $\varepsilon_{90\%}$ with s_p in the different sub-planes (left y-axis for the sub-plane Q_1 , right y-axis for the sub-plane Q_2) and with the different pointing strategies.

Fig. 3. Incidence of the choice of the optical source orientations on the positioning performance.

towards the points $C_p = O_2 + A_p$ and $D_p = O_2 + B_p$. The resulting evolution of the RMS error $\varepsilon_{90\%}$ with s_p is shown in Fig. 3(b) (orange curve with the circular markers, right y-axis) and allows to conclude that these new sets of orientations do enable better positioning, with $\varepsilon_{90\%}$ reaching at best 16.9 cm.

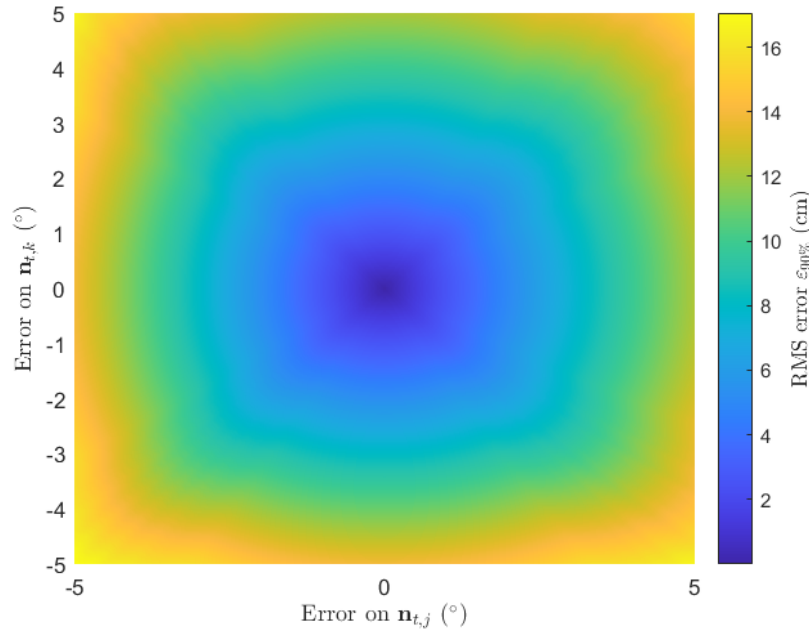


Fig. 4. Evolution of the RMS error $\varepsilon_{90\%}$ with the pointing errors on $\mathbf{n}_{t,j}$ and $\mathbf{n}_{t,k}$.

Despite this almost twofold increase in performance, the RMS error remains rather larger over Q_2 . Given that AWGN is here neglected, this error therefore might be due to larger NLOS contributions on the received optical power than over Q_1 . To verify this hypothesis, we reproduced the previous simulations, this time considering an $8 \times 8 \times 2.5$ m³ room, i.e. pushing the walls away from the Q_2 sub-plane. The resulting RMS error, in orange with star markers in Fig. 3(b), is in this case lower indeed, reaching 11.7 cm when $s_p = 2$ m. Note that with $s_p = 4$ m, $\varepsilon_{90\%}$ even goes down to 5.4 cm. These latter results therefore confirm the interest of the proposed positioning method, not only for areas of interest close to the source used like Q_1 , but also for more distant areas, as long as they are not close to over-reflective obstacles like walls.

These results assume, however, that the various source orientations are perfectly known, a condition that may not be verified in practice. Figure 4 shows the evolution of the error $\varepsilon_{90\%}$ over the entire receiver plane without considering AWGN and reflections, and as a function of variations in the orientations vectors $\mathbf{n}_{t,j}$ and $\mathbf{n}_{t,k}$ through variations in the orientation angle ω . A 1° error in knowledge of ω therefore results in an increase of 3.4 cm in $\varepsilon_{90\%}$, a value that can go up to 16.7 cm in the case of 5° error in ω . Good knowledge of these orientations is therefore fundamental to maintaining high positioning performance.

4. Experimental proof-of-concept

A first proof-of-concept, described in Section 4.1, has been implemented to test and partially validate experimentally the proposed positioning technique. We focused on two main objectives: 1) to demonstrate the feasibility of positioning on one dimension, i.e. along an axis, as described in Section 4.2, and 2) to verify that our positioning method does not depend on the orientation of the receiver, as detailed in Section 4.3.

4.1. Proof-of-concept description

The test bench used to validate the positioning method proposed here is shown in Fig. 5. On the AP side, the optical source used is an Osram LZ1-00R702 dual junction infrared LED ①, driven at 300 mA and operating at 940 nm. It is installed on a homemade mechanical stage ②, itself fixed to a bracket ③ at 2.23 m height. This stage is designed to control the orientation of the LED's optical axis with two rotational degrees of freedom around the $\mathbf{u}_x$ and the $\mathbf{u}_y$ axis. It consists of two orthogonal Thorlabs PRMTZ8/M motorized precision rotation stages ④, controlled themselves by a computer ⑤ via Thorlabs KDC101 servo motor controllers ⑥. As the LED is located at the center of rotation of this mechanical stage, any change in its orientation does not change its position. On the UE side, the receiver is a homemade opto-electronic front-end composed of an Osram BPX61 silicon photodiode and a transimpedance stage ⑦, which output voltage is recorded by a Keithley 2100 high-precision voltmeter ⑧ and then post-processed with the computer ⑤ to estimate the receiver's location. This front-end is attached to the gripper of a Universal Robots URS robot arm ⑨, which position and orientation can be controlled by the computer ⑤. In this section, we will consider that the height of the receiver remains fixed, at 90 cm from the ground (i.e. $\mathbf{R} = [x, y, -1.33 \text{ m}]$).

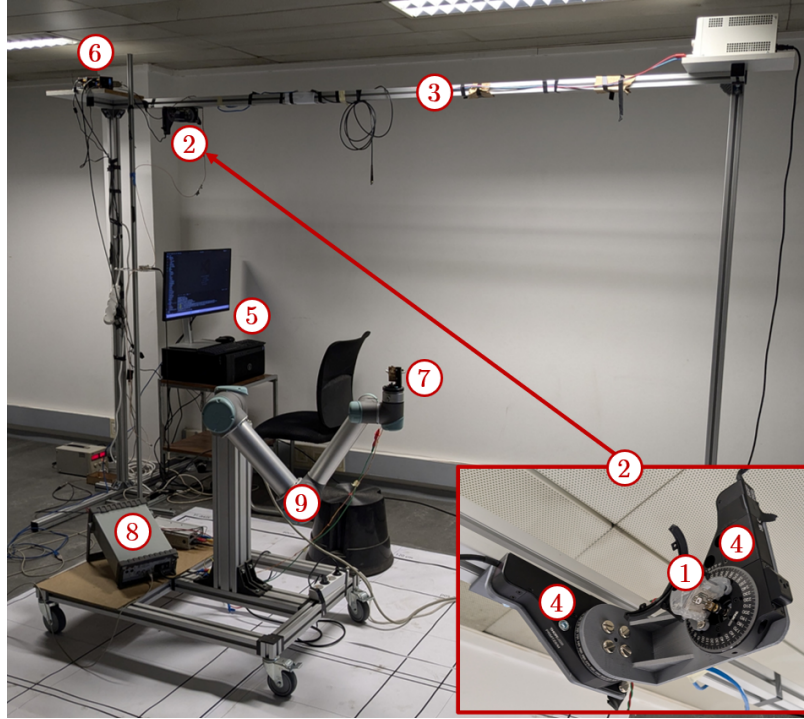


Fig. 5. Test bench used to validate the proposed positioning method.

As mentioned in Section 2.2, the positioning method we propose here assumes that the optical source used has a Lambertian beam pattern following a generalized cosine law of order m_l . Here, the theoretical semi-angle of emission of the LZ1-00R702 LED is 45° , so that $m_l = 2$. However, it is necessary to check whether this pattern is indeed Lambertian. For this, the source was first directed toward the ground ($\mathbf{T} = [0, 0, 0]$, $\mathbf{n}_{l,i} = [0, 0, -1]$) whereas the photoreceiver was placed directly beneath the source and pointing toward it ($\mathbf{R} = [0, 0, -1.33 \text{ m}]$, $\mathbf{n}_r = [0, 0, 1]$), so that a reference RMS voltage could be recorded. The angle of orientation of the source ω , i.e. in this case the angle of emission $\phi_{l,i}$, was then varied from -90° to 90° around the $\mathbf{u}_x$ axis and the

resulting received RMS voltage was again recorded and normalized with the reference RMS voltage, so the emission pattern shown in Fig. 6 could be plotted.

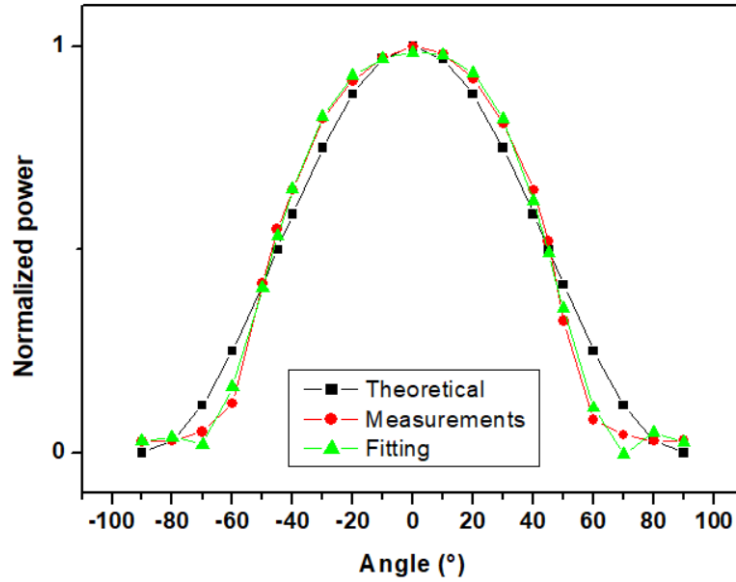


Fig. 6. Emission pattern of the LZ1-00R702 LED source with a theoretical semi-angle at half-power of 45° .

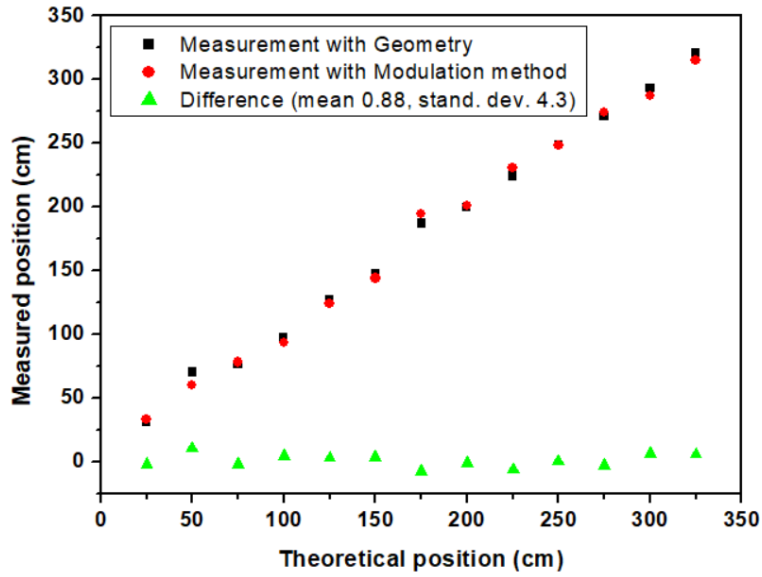
As we can see, as far as the angle of emission remains below $\pm 60^\circ$, the Lambertian model, in black, is quite close to the measurements, in red, although both curves do not match exactly. These differences could therefore lead to positioning errors, through variations in the K_{ij} factor, as indicated in Section 2.3. Note, however, that if necessary, the real pattern can be approximated by an eight order polynomial function, shown in green, which could be used in (10) and (11) to replace the Lambertian model.

4.2. One-dimensional positioning method validation

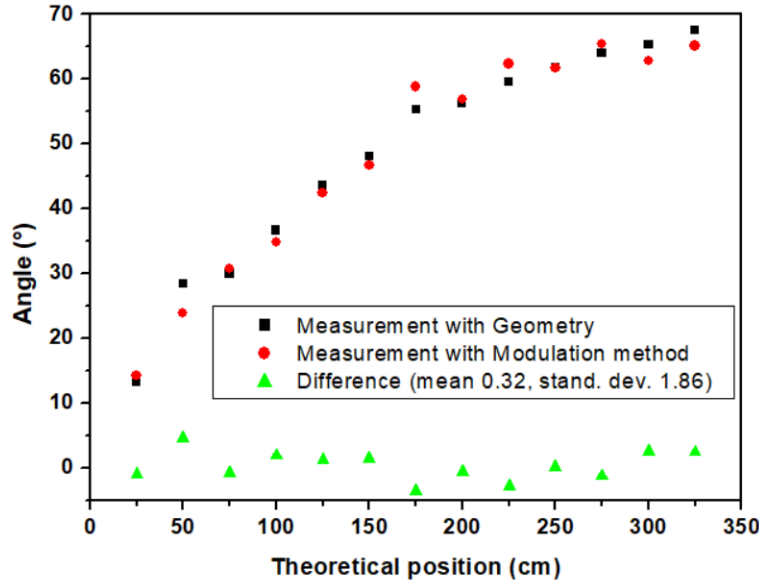
With this experimental platform, we first checked whether one-dimensional positioning was possible. To that end, we directed the source toward the ground ($\mathbf{n}_{t,i} = [0, 0, -1]$) and placed the receiver at a given ‘theoretical position’ $\mathbf{R} = [x, 0, -1.33 \text{ m}]$ along the $\mathbf{u}_x$ axis, measured from landmarks on the ground, while orientating it toward the source manually to increase the collected optical power. Note that since this alignment process was manual, the angle of reception ψ_r was most probably not null but, as we will see in Section 4.3, the position measurement is insensitive to this angle.

We then took a first set of measurements to determine the ‘Geometric position’ of the receiver, then used as the ground truth. To do this, we measured with a laser rangefinder the coordinates of the source and receiver along the $\mathbf{u}_z$ axis, as well as the distance d separating them. These measurements then enabled us to determine the position of the receiver along the $\mathbf{u}_x$ axis as well as the corresponding angle of emission $\phi_{t,i}$ using simple trigonometric relations. Note that the resolution of this reference measurement is 1 mm but the positioning uncertainty of the rangefinder does not allow us a 1 mm accuracy, which we estimate to be less than 5 mm instead.

After this first step, we took a second set of five measurements, this time to estimate the position of the receiver using our positioning method. First, we measured the reference voltage resulting from background lighting while the LED was off. This background level was then subtracted to the next measurements to remove this noise source. Once this reference was obtained, the



(a) Evolution, with the 'Theoretical position', of the 'Geometric position' and 'Estimated position' along with their difference.



(b) Evolution, with the 'Theoretical position' of the receiver, of the angles of emission $\phi_{t,i}$ of the source as estimated with the geometric measurements and the proposed positioning method, along with their difference.

Fig. 7. Experimental results for the measurements with a single rotating source. For both curves, each step represents a displacement of 25 cm.

LED was switched on and oriented successively in four different directions by rotations of an angle $\omega = 0^\circ, 30^\circ, 45^\circ$, and 60° around the $\mathbf{u}_y$ axis. For each orientation, the RMS voltage of the received signal was recorded, so that six ratios of type K_{ij} could be calculated. These ratios

were eventually used to get, on the one hand, several estimates of x using (17), which were then averaged to get the final 'Estimated position', and on the other hand, an estimate of the angle of emission $\phi_{t,i}$.

This whole process was eventually repeated for several positions x of the receiver, ranging from 0 m to 3.25 m by steps of 25 cm, so that Fig. 7 could be obtained. In Fig. 7(a), the 'Geometric position' and the 'Estimated position' are plotted respectively in black and red against the 'Theoretical position', along with their difference, in green. We can see that this difference is quite low, with a mean value of 0.88 cm and a standard deviation of 4.3 cm. At the same time, Fig. 7(b) shows the evolution, with the 'Theoretical position', of the angle of emission $\phi_{t,i}$ measured with the geometric approach, in black, and estimated with our method, in red. Here, again, the difference between both curves, in green, has a low mean value of 0.32° and a standard deviation of 1.86° . These results therefore validate the capability of the proposed system to provide accurate positioning in one dimension.

Note that, in practice, the time required to calculate the positioning variables – which are obtained after a few basic operations, as reminded by (17) and (18) – is of the order of a millisecond. However, the transition time to move the AP from one orientation to the next is of the order of a second, because the Thorlabs PRMTZ8/M motorized rotation stages were primarily chosen for their ease of use and high accuracy, rather than for their speed. As it stands, the prototype described here would therefore be impractical for tracking moving targets, but this limitation stems above all from the optical beamsteering technique used and not from our positioning algorithm, which would be perfectly suitable for such applications when coupled with much faster beamsteering solutions, such as phased arrays or intelligent reflecting surfaces [2].

4.3. Sensitivity to receiver orientation

In a second step, we checked whether the proposed method is indeed insensitive to the orientation of the receiver, which is another important feature of our positioning technique. To do this, we

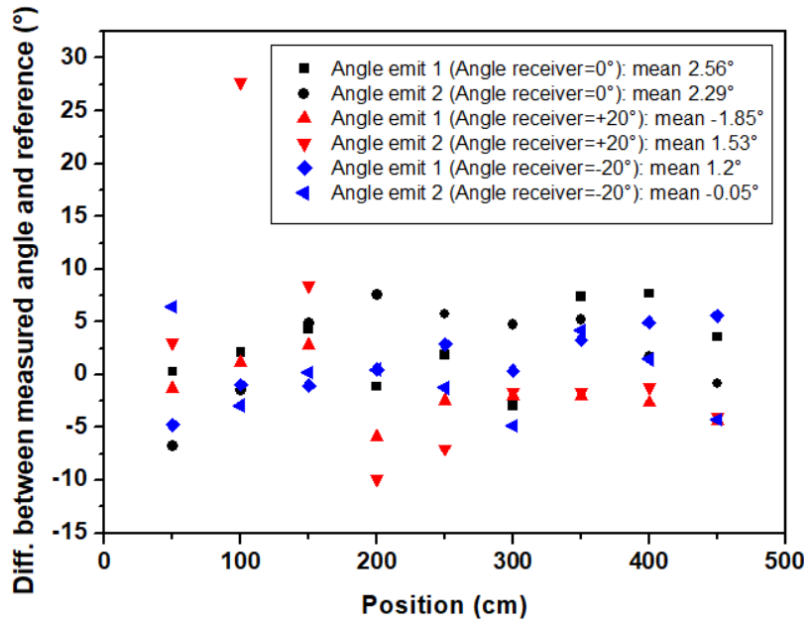


Fig. 8. Evolution, with the position of the receiver along the $\mathbf{u}_x$ axis, of the difference between the reference and estimated angles of emission $\phi_{t,i}$ for both transmitters $\mathbf{T}_1$ and $\mathbf{T}_2$.

used two emitters placed respectively at coordinates $\mathbf{T}_1 = [0, 0, 0]$ and $\mathbf{T}_2 = [5 \text{ m}, 0, 0]$ and at a 2.23 m height. The receiver, still located at a 90 cm height, was then moved from the point directly below $\mathbf{T}_1$ to the point directly below $\mathbf{T}_2$ along the $\mathbf{u}_x$ in 50 cm steps, while first keeping its orientation toward the ceiling, i.e. $\mathbf{n}_r = [0, 0, 1]$.

For each position, the angle of emission $\phi_{t,i}$ of both sources was first calculated using rangefinder measurements similar to those presented in the Section 4.2 to get a reference, and then using the proposed positioning method. The difference between these measurements was then plotted against the 'Theoretical position' of the receiver, as shown in Fig. 8, where the black squares give the difference in $\phi_{t,i}$ in the case of $\mathbf{T}_1$ and the black dots this difference in the case of $\mathbf{T}_2$.

This whole process was then repeated with the receiver oriented at an angle of $\pm 20^\circ$ around the $\mathbf{u}_y$ axis. The resulting differences between the reference angle of emission and the angle of emission estimated with our method are plotted in Fig. 8, respectively in red and blue for the $+20^\circ$ and -20° cases. One can see that except a point at 1 m, where the experiment may have been disturbed, all the measurements remain in the same range of a few degrees range whatever the angle of reception, which confirms the proposed positioning method does not depend on the orientation of the receiver.

5. Conclusions and future works

In this paper, we propose a new positioning method for OWC based on the possibility of modulating the orientation of optical sources using beamsteering techniques and combining the principles of AOA and RSS to estimate the position of a receiver in a 2D plane. Compared with existing positioning techniques, our method requires only a single source and a single photoreceiver, and neither additional sensors nor knowledge of the emitted power. It is also not sensitive to receiver orientation, and is not limited to optical sources, and could be used with other types of antennas, such as RF antennas.

The simulation analysis proposed in Section 3 shows that this method can achieve 2D positioning errors of less than 10 cm, despite AWGN and reflections, provided the source orientations are carefully chosen, and even if the receiver is not in the immediate vicinity of the source. On its side, the experimental proof-of-concept study detailed in the Section 4 validates the operation and good performance of this method for one-dimensional positioning, as well as its non-sensitivity to receiver orientation.

This experimental validation would obviously need to be supplemented by 2D positioning tests. Although results have already been obtained on this subject, they are not detailed here, as the primary intention of this article is to present in detail the fundamental principles of this method. These results, which will be reported in future work, nevertheless show that a positioning error of less than 5 cm can be achieved by coupling our method with machine learning techniques.

In addition to exploiting these results, our future work will involve improving our method to enable three-dimensional positioning in a dynamic environment and to improve its practicability using more advanced beamsteering solutions such as optical phased arrays or intelligent reflecting surfaces. Finally, given the sensitivity of the positioning error to the choice of source orientations, we intend to focus our future efforts on defining optimal strategies for defining such orientations, which will require an improved modeling of the system and the development of more advanced positioning estimators, including using machine learning.

In any case, the method proposed here is to our knowledge the first solution to exploit optical beamsteering techniques to enable positioning, and could therefore be integrated into any OWC system employing these techniques to implement OWC-based ISAC solutions for applications ranging from Industry 4.0 to satellite-to-satellite communications.

Appendix: System non-solubility with unknown x , y and z

Let the system of equations described by (15) and (16). The goal is here to evaluate the determinant of $\mathbf{A}$, which can be done by building $\mathbf{A}$ with the vectors of the canonical basis $\mathcal{B} = \{\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z\}$ of the main frame $\mathcal{F} = (T, \mathcal{B})$ instead of random orientations. From a purely mathematical point of view, choosing these three different orientations as the vectors of $\mathcal{B}$ does not introduce any limitations, as any vector in $\mathbb{R}^3$ can be expressed as a linear combination of $\mathbf{u}_x$, $\mathbf{u}_y$ and $\mathbf{u}_z$, so the resulting matrix $\mathbf{A}$ would have the same determinant as the one obtained with the canonical basis.

Knowing that $\mathbf{u}_x = [1, 0, 0]$, $\mathbf{u}_y = [0, 1, 0]$ and $\mathbf{u}_z = [0, 0, 1]$, and assuming $\mathbf{n}_{t,i} = \mathbf{u}_x$, $\mathbf{n}_{t,j} = \mathbf{u}_y$ and $\mathbf{n}_{t,k} = \mathbf{u}_z$, we can rewrite $\mathbf{A}$ from (16) as:

$$\mathbf{A} = \begin{bmatrix} 1 & -K_{ij} & 0 \\ 0 & 1 & -K_{jk} \\ 1 & 0 & -K_{ik} \end{bmatrix}, \quad (26)$$

which determinant is:

$$\begin{aligned} \begin{vmatrix} 1 & -K_{ij} & 0 \\ 0 & 1 & -K_{jk} \\ 1 & 0 & -K_{ik} \end{vmatrix} &= -K_{ik} + K_{ij}K_{jk} \\ \iff |\cdot| &= -\left(\frac{\cos(\phi_{t,i})}{\cos(\phi_{t,k})}\right)^{\frac{1}{m_i}} + \left(\frac{\cos(\phi_{t,i})}{\cos(\phi_{t,j})}\right)^{\frac{1}{m_i}} \left(\frac{\cos(\phi_{t,j})}{\cos(\phi_{t,k})}\right)^{\frac{1}{m_j}} \\ \iff |\cdot| &= \left(\frac{\cos(\phi_{t,i})}{\cos(\phi_{t,k})}\right)^{\frac{1}{m_i}} (-1 + 1) = 0. \end{aligned} \quad (27)$$

We can therefore deduce that as such, the system (15) with unknowns x , y and z and matrix $\mathbf{A}$ defined by (16) has no unique solution.

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